

Control systems

Motor gear compressor

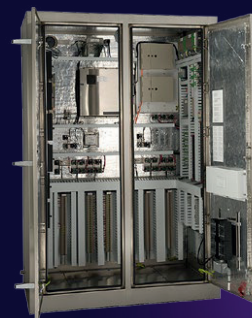
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Siemens Energy has been designing control systems for various applications for more than 50 years. With this heritage comes a wealth of experience designed into every system. The motor gear compressor (MGC) control system is an integrated solution designed to monitor vibration, temperature, surge control, and other process variables to safely and efficiently operate an advanced motor gear-driven centrifugal compressor train. These latest systems use the most up-to-date, state-of-the-art technology as described below.

At the heart of the system is a programmable logic controller (PLC). Today's PLCs are technically advanced, flexible in configuration and programmed in IEC61131-3 languages such as Structured Text, Ladder Diagram, Function Block Diagram and Sequential Function Charts, the process control programs are ready-to-use with comprehensive documentation. Surge control and other high-level programs are developed in dedicated and segregated blocks to ensure the highest levels of system reliability and security.

The PLC-based controllers interface with the instrumentation of the motor gear VFD compressor through an input/output (I/O) system and communicate both internally and externally through standard industry protocols.

These benefits translate into lower life cycle costs associated with spare parts, ease of operation, reduced maintenance, and comprehensive training as well as improvement in performance and throughput. In addition, the use of advanced packaging techniques provides easy access to panel internal components with the smaller panels offering a distinct advantage where space is limited.



To safely and efficiently operate a motor gear centrifugal compressor train, the control system must be technologically advanced and capable of meeting the operational challenges found in today's demanding industrial environments.

System configuration and benefits

The MGC control system free-standing enclosure is based on either Allen Bradley, GE or Triconex PLC controllers (different brands available), and incorporate the following standard functions:

- Capacity/performance control system
- Thrust bearing axial position and temperature monitoring
- Bearing vibration and temperature monitoring OneControl, patented
- Anti-surge control
- Pressure monitoring
- Process and auxiliaries temperature monitoring
- Oil system monitoring
- Start, stop, alarm, and shutdown sequencing

System communication

Communication to external system devices is provided using the MODBUS TCP/IP protocol. Other protocols are available upon request.

Hard-wired relay backup shutdown system

An independent hard-wired relay back-up system is installed to shut down the control system in critical situations including PLC failure, vibrations exceeding designated limits and other critical system inputs.

Standard system components

The following components and displays are provided as part of the core system:

- 17" touch screen monitor (D-R HMI – panel mounted)

Indicator lights:

- Permissive to start – Green
- Unit running – Green
- Common alarm – Amber
- Common shut down – Red
- PLC failure – Red
- Normal stop – Blue

Pushbuttons:

- Unit start
- Unit normal stop
- Alarm silence
- Acknowledge
- Reset
- Speed raise*
- Speed lower*
- ESD – pull to stop (no guard)

Selector switches:

- Mode select (local/remote)
- Speed controls (auto/manual)*
- External light switch**

System options and enhancements

The following options are available and can be added upon request:

- Condition monitoring system (see separate literature)
- Additional operator pushbuttons and/or pilot lights
- Non-standard communications protocols
- RS232/485 serial communications
- DCS communications
- Hazardous area panel classification and certification
- Laptop computer (programming and maintenance device)
- Additional local operator panel
- Non-standard HMI systems (Different brands as replacement)
- Monitoring for VFD motor driver
- Intrinsically safe barriers or isolators
- Additional anti-surge loops
- Additional compressor casings
- Upgrade to redundant configuration
- Client-witnessed factory acceptance test

* Required for VFD applications along with dedicated speed meter

** LCP applications

Published by
Siemens Energy Global GmbH & Co. KG
Transformation of Industry
Siemenspromenade 9
91058 Erlangen
Germany

Published by
Siemens Energy, Inc.
Transformation of Industry
4400 N Alafaya Trail
Orlando, FL 32826
USA

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